



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PURE MATHEMATICS

PAPER 1

6042/1

3 hours

NOVEMBER 2025 SESSION

Additional materials:

Answer booklet

Graph paper

List of Formulae MF7

Scientific calculator [Non-Programmable]

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

Answer **all** questions.

This question paper consists of 5 printed pages.

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[Turn over



- 1 The mass, m , of a soccer ball made of plastic of uniform density varies directly as the cube of its diameter, d . A soccer ball with diameter 7 cm has a mass of 0.11 kg.

Calculate the mass of a similar soccer ball of diameter 9 cm.

[4]

- 2 Given the differential equation

$$\frac{dy}{d\theta} + 2\sin 2\theta = 0, \quad \text{where } y = 1, \quad \text{when } \theta = \frac{\pi}{6},$$

find the equation connecting y and θ .

[4]

- 3 Solve the equation $z^2 + 3z + 4i + 2 = 0$, giving the answer in the form $x + iy$, where x and y are real.

[5]

- 4 (a) Find, in terms of θ , an appropriate value of

$$\frac{\sin 4\theta + \tan 2\theta}{3 + \cos 2\theta},$$

for small values of θ , rad ($\theta \rightarrow 0$), neglecting terms in θ^2 and higher powers.

[2]

- (b) Given that $1^\circ \approx 0.018$ radians and $(0.018)^2 \approx 0.000\,324$, find an approximate value for $\frac{\cos 2^\circ}{\sin 1^\circ}$.

Give the answer correct to 2 decimal places.

[3]

- 5 Given that the equation

$$x^2 + 2x + 4 = P(x^2 + 4)$$

has equal roots, find the possible values of P .

[5]

- 6 Prove that the transformation matrix $\begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$ maps all points on the line $y = 3x$ onto themselves.

[5]

- 7 Solve the equation

$$2(3^{2x+1}) - 20(3^x) = -6$$

[6]

- 8 Prove by mathematical induction that

$$\frac{1}{2} + \frac{1}{6} + \dots + \frac{1}{n(n-1)} = 1 - \frac{1}{n} \text{ for } n > 1.$$

[6]



- 9 The line LM satisfies the equation $2x - 14y + 6 = 0$. Points L and M have coordinates $(-3; 0)$ and $(k; 1)$ respectively.

Find the

- (a) value of k . [2]
- (b) length of LM, leaving your answer in surd form. [2]
- (c) equation of the line through M, which is parallel to the line $2x + y + 5 = 0$. [3]

- 10 Given that the equation of a curve is

$$y = 10x - x^2 - 27,$$

- (a) express $10x - x^2 - 27$ in the form $-(a - x)^2 + b$, where a and b are integers. [2]
- (b) Hence or otherwise show that $10x - x^2 - 27$ is always negative. [1]
- (c) State the coordinates of the maximum point of the curve $y = 10x - x^2 - 27$. [2]
- (d) Sketch the curve, showing the point where it crosses the y -axis. [3]

- 11 Given $f(x) = \frac{2x+5}{x+1}$, $x \in \mathbb{R}$, $x \neq -1$,

- (a) show that $f(x) \equiv 2 + \frac{3}{x+1}$. [2]
- (b) describe the sequence of transformations that map the graph of $y = \frac{1}{x}$ onto the graph of $y = f(x)$. [3]
- (c) sketch the graph of $y = f(x)$ for $x \geq -1$. [3]

- 12 By sketching suitable graphs, show that the equation

$$\cot x = 2x^2 - 1$$

has only one root in the interval $0 < x < \frac{\pi}{2}$. [3]

- 13 Given that $(G, *) = \{0; 1; 2; 3\}$.

Show whether $(G, *)$ is a group or not under addition modulo 4. [9]



- 14 Functions f and g are defined by

$$f: x \rightarrow e^{3x}, x \in \mathbb{R}$$

$$g: x \rightarrow \ln(x + 2), x \in \mathbb{R}, x > -2.$$

- (a) Express in terms of x ,
- $f^{-1}(x)$, stating the domain. [6]
 - $fg(x)$. [3]
- (b) Sketch on a single diagram, the graphs of $y = f(x)$ and $y = f^{-1}(x)$. [3]
- 15 (a) By sketching suitable graphs, on the same axes, show that the equation
- $$e^{2x} + 4x - 5 = 0, \text{ has exactly one root.} \quad [3]$$
- (b) Show by calculation that the root of the equation in (a) lies between 0 and 1. [3]
- (c) Use the Newton-Raphson method, starting with the value $x_0 = 0.5$, to find the root of
- $$e^{2x} + 4x - 5 = 0,$$
- correct to 2 decimal places. [5]
- 16 (a) Find the stationary points on the curve
- $$y = 2x^3 + 3x^2 - 12x + 1$$
- and determine their nature. [9]
- (b) Hence sketch the curve in (a) showing clearly the position of the stationary points. [3]



17 It is given that $P = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & -2 & 3 \end{pmatrix}$.

(a) Find the inverse of P.

[7]

(b) Hence or otherwise solve the following equations

$$x + y + 2z = 0$$

$$2x - y + z = -3$$

$$x - 2y + 3z = 5$$

[6]

